

# Thomas Madden

La Jolla, CA | [thmadden@ucsd.edu](mailto:thmadden@ucsd.edu) | +1 (510) 220-9197 | [LinkedIn](#)

## Education

University of California, San Diego, B.S. in Mathematics

September 2023 – June 2027

- GPA: 3.89/4.0; Provost Honors all quarters

## Research Experience

UC San Diego Mathematics Honors Program

September 2026 - Present

- Acceleration for Randomized Kaczmarz and randomized low-rank approximation for scientific computing.
- Advised by Robert J. Webber

UCLA Computational and Applied Mathematics (CAM) REU

June 2026 – August 2026

- Proved sharp convergence guarantees for Fisher-Rao and Wasserstein-Fisher-Rao gradient flows
- Identified properties of initializations which lead to better performance of algorithms based on these flows for sampling from mixture distributions
- Clarified necessary and sufficient conditions for convergence of the flows
- Advised by Yifan Chen

Scalable Optimization and Control (SOC) Lab at UC San Diego

May 2025 – Present

- Introduced algorithm that generically accelerates the proximal bundle method in convex optimization
- Showed that incorporating iterate-acceptance procedure allows large stepsize selection
- Performed numerical simulations using JAX, CVXPY, and NumPy to illustrate speed-up
- Advised by Yang Zheng

Random matrices and  $\Lambda$ -distributions, Polymath Jr.

June 2025 – September 2025

- Proved exact criterion for independence of classical random variables obtained from  $\Lambda$ -random variables, with applications in random matrix statistics and number theory
- Presented results on behalf of 25+ person group at end-of-project conference
- Advised by Sean Howe

## Papers

### Published:

1. *An accelerated proximal bundle method for convex optimization*  
F.-Y. Liao, T. Madden, and Y. Zheng. *Learning for Dynamics and Control (L4DC 2026)*, PMLR 331:1012–1034, 2026. <https://proceedings.mlr.press/v331/liao26a.html>

### Preprints:

1. *On the convergence of Fisher-Rao and Wasserstein-Fisher-Rao gradient flows of the KL divergence*  
T. Madden and Y. Chen. *In preparation*.
2. *Matrix group  $\Lambda$ -distributions*  
M. Bertucci, S. Van Blarcom, V. Glancy, S. Howe, T. Madden, B. Miller, S. Pal, G. Papagrigoriou, N. Raikman, and T. Tran. *Submitted, 2026*. <https://arxiv.org/abs/2608.16796>

## Presentations

### Talks:

**August 12, 2026** "On the convergence of Fisher-Rao and Wasserstein-Fisher-Rao gradient flows of the KL divergence," UCLA Computational and Applied Mathematics (CAM) REU Research Presentation

**February 13, 2026** "Acceleration with large gradient steps via the proximal bundle method," UC San Diego Mathematics of Information, Data, and Signals Seminar.

**January 7, 2026** “Random matrices and  $\Lambda$ -distributions,” AMS Special Session on Polymath Jr. Student Research.

## Posters:

**May 28, 2026** “Acceleration with large gradient steps via the proximal bundle method,” UC San Diego Undergraduate Engineering Research Symposium (UERS).

## Relevant Coursework

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- MATH 140ABC: Real Analysis
  - MATH 100ABC: Abstract Algebra
  - MATH 148: Analysis of Partial Differential Equations
  - MATH 180A: Introduction to Probability
  - MATH 180B: Introduction to Stochastic Processes I
  - MATH 114: Computational Stochastics
  - MATH 154: Graph Theory
  - MATH 120A: Introduction to Complex Analysis
  - MATH 105: Basic Number Theory
  - MATH 240A: Real Analysis (*i.p.*)\*
  - MATH 286: Stochastic Differential Equations (*i.p.*)\*
  - MATH 280ABC: Probability Theory
  - MATH 281A: Mathematical Statistics
  - MATH 216A: Supervised and Unsupervised Learning
  - MATH 216B: Deep Learning
  - MATH 216C: High-Dimensional Probability
  - MATH 294: The Mathematics of Finance
  - DSC 291A: Stochastic Optimization
  - DSC 291A: Sampling and Diffusion Models
- \*in progress abbreviated as i.p.*

## Teaching Experience

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**Content Tutor**, Teaching + Learning Commons at UC San Diego September 2024 – June 2025

- Tutored undergraduates in all lower division calculus courses, linear algebra, and mathematical reasoning
- Led sessions with 1-3 students from different classes attending simultaneously
- Incorporated abstract motivation for concepts in calculus, along with physical intuition
- Received 100% recommendation rating from students

**Course Facilitator**, Subject Inc. February 2024 – September 2024

- Graded assignments and provided individualized feedback for 300+ students in Algebra II and Geometry
- Made 100+ adjustments to curriculums to improve clarity/correctness of solution manuals
- Tutored students as needed to maintain ~90% improvement in mathematical aptitude

## Activities

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**Society of Undergraduate Mathematics Students**, President January 2025 - Present

- Created Pi Mu Epsilon chapter at UC San Diego to provide scholarship and conference travel funding opportunities to undergraduates engaged in research
- Planned math-related workshops, including professional development and networking opportunities
- Hosted annual integration bee with original and challenging problems

**Society of Undergraduate Mathematics Students Reading Group**, Lead December 2025 - Present

- Lead a graduate-level reading group on "Statistical Optimal Transport" by Sinho Chewi, Jonathan Niles-Weed, and Philippe Rigollet

## Additional

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**Libraries:** JAX, CVXPY, NumPy

**Languages:** English (native) and Spanish (fluent)

**Awards:** 2024 UCSD Integration Bee Winner; Received highest score school-wide on Integration Bee Qualifying Exam, HSF Scholar